

Balancing Human Agency and AI Autonomy in Human-AI Idea Selection

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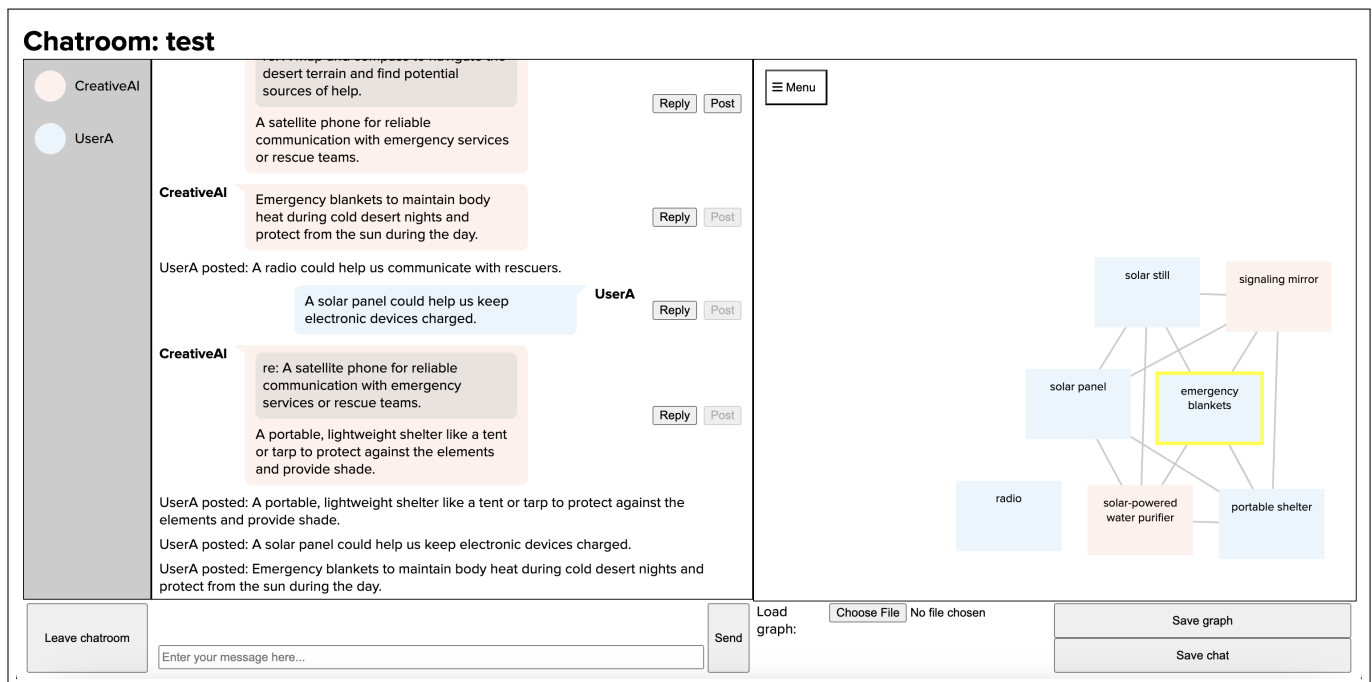


Figure 1: IdeaMap human-AI group brainstorming system, featuring a sidebar with all usernames and their associated colors (left), a chatroom for communicating and sharing ideas (middle), and an ideation map visualization based on posted ideas (right).

Abstract

Many generative AI systems are built for domain-specific knowledge work. These tasks often involve idea generation with verifiable, clearly defined outputs. The subjective nature of evaluating creative outputs, however, has traditionally involved final idea curation by humans. Extending this human-only idea selection process to

human-AI teams has the potential to boost performance by leveraging AI capabilities, but how to best balance human agency and AI autonomy remains unclear. We used a technology probe, IdeaMap, with a chatroom and ideation map visualization to elicit feedback on desired AI features for convergent thinking (idea selection) in single and multi-party group brainstorming in a participatory design workshop (12 participants). We uncovered user perceptions on human agency and AI autonomy in human-AI teaming for convergent thinking, and discuss how to balance them with a situationally aware, flexible AI that responds to both explicit user feedback and observed user behaviors.

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CCS Concepts

• **Human-centered computing** → **Computer supported cooperative work**; *Visualization systems and tools*.

Keywords

human-AI collaboration, human agency, AI autonomy, idea mapping

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1 Introduction

Recent AI advancements have shifted the paradigm of “AI as tool” to “AI as partner” and currently to AI as agent [9, 18]. Generative AI models, such as LLMs, can quickly produce many outputs for human-AI applications ranging from text chatbots [17] to human-AI decision making [14]. Many of these AI applications are geared towards knowledge work with a specific idea generation and/or idea selection task, evaluated via verifiable “correct” outputs.

Shneiderman’s human-centered AI (HCAI) framework suggests a two-dimensional model for human agency and AI autonomy at the broader task level, advocating for Reliable, Safe, and Trustworthy (RST) applications [25] by balancing human control and system automation. Muller and Weisz extend Shneiderman’s RST framework to multi-party human-AI teams and introduce dynamic, shifting levels of human and machine initiative between each step in a task [19]. However, Muller and Weisz don’t examine the more minute nuances of individual behaviors in teams comprised of equal roles and responsibilities: in their model, levels of human and AI initiative for a given step remain fixed. When humans’ performance varies throughout the execution of a given step in collaborative work, however, there could be more nuanced ways to combine and integrate human control and AI autonomy. It is unclear why levels of human control and AI autonomy should remain fixed.

In creative applications, idea generation and selection are sequentially related to one another, and equally important. The conventional design thinking process for brainstorming ideas for design and options for decision-making typically features two phases of cognitive work: divergent thinking (idea generation or producing a large quantity of ideas on a given problem) followed by convergent thinking (idea selection or selecting a smaller set of ideas from the pool by considering the relevance and quality of ideas) [4, 5]. Existing human-AI co-creation systems where users operate AI tools to generate content in a range of mediums from writing text [27] to digital image generation [8] tend to put more emphasis on quantity, or using AI to boost idea generation productivity. The curation of quality content—the idea selection task—tends to be overlooked in human-AI designs, as users explore and curate the final results on their own without deliberate support. While AI has successfully contributed to these creative generative results, the role of AI and the optimal levels of autonomy when working with humans on idea selection remains unclear.

We explore how users perceive human agency and AI autonomy in human-AI group brainstorming, specifically convergent thinking, through a technology probe study [12] using an early-stage design prototype, IdeaMap. IdeaMap is a human-AI brainstorming system that uses a LLM (gpt-3.5-turbo) with a chatroom and ideation map visualization that functions as a shared knowledge workspace. We sought answers to the following research questions with IdeaMap: **RQ1.** How do users perceive their control according to their beliefs and perceptions of what AI can do, and how AI functions in human-AI teaming for convergent thinking?

RQ2. How do we design to balance user-control (agency) and AI (autonomy) over a shared knowledge workspace, specifically in convergent thinking, such as collaborative selection of ideas?

We conducted the technology probe study with two types of human-AI groups through a participatory design session: single (1 human, 1 AI), and multi-party (2 humans, 1 AI) to elicit users’ perceptions of AI involvement in idea selection for diverse group settings. We elicited user perceptions on human agency and AI autonomy in human-AI teaming for convergent thinking, and discuss the potential of a situationally aware, flexible AI to accommodate variances in user behavior within and between steps of a task.

2 Background

2.1 Human Agency and AI Autonomy

Existing work on human-AI interactions offer a general framework concerning human agency and AI autonomy that may jointly influence interaction outcomes, such as task productivity or creativity in knowledge work [19, 25]. These frameworks, however, operate at the broader levels of use cases and system configurations, which tend to overlook nuanced experiences and perceptions that human participants may have in human-AI teaming scenarios.

Different levels and combinations of human agency and AI autonomy can affect human-AI teaming outcomes in sophisticated means. While high user agency supports dynamic interactions, structured workflows guide tasks more efficiently [10, 22]. In addition, emphasizing user control in co-creation tasks can lead to better engagement and satisfaction [18]. Through increasing levels of AI capabilities and autonomy, the usage of AI has transitioned from tools to collaborative “co-pilot” partners [9] in many domains ranging from decision-making [14] to code development [17]. When AI isn’t conceptualized and operationalized as a tool, but a collaborating partner, it raises concerns whether AI autonomy may start replacing human agency and how the replacement or displacement of human agency may impact not only the work, but also the human workers. Some work has shown that an adaptive autonomy approach, where AI adjusts its autonomous behaviors based on the collaborative context, can further improve team performance and user satisfaction over a fixed or full level of autonomy [18, 22].

In the context of knowledge work, forming human-AI teams with both human and AI peer actors to produce and select creative ideas has potential to transform the process and outcomes of group creativity. AI tools, such as LLMs, may potentially help groups generate more diverse, innovative ideas and improve team dynamics during group brainstorming [23, 24]. Initial work has focused on designing AI tools to enhance multi-party brainstorming [23, 24] and group decision-making [20, 31]. In some initial attempts of

human-AI teaming, users appreciate AI's efficiency and inspiration, but prefer human collaborators' originality and trust [11]. One observation is that involving AI actors could possibly affect group dynamics by reinforcing values such as objectivity and fairness, but struggle with incorporating human opinions, dynamically engaging in discussions, and finding synergies in the group [20, 31]. Therefore, in our work, we probe the balance of human agency and AI autonomy to explore designing support for human-AI teaming in collaborative knowledge work, especially in the process of creative idea selection.

2.2 Human-AI Co-Creation Systems

Human-AI co-creation systems have enabled task-specific design space explorations from writing short stories [27] to creating news reels [30]. These human-in-the-loop workflows encourage sequential divergent and convergent thinking throughout the human-AI co-creation process [8, 27, 28]. While human-in-the-loop co-creation workflows with sequential turn-taking can retain users' perceived agency [13], other human-AI co-creation methods feature higher AI autonomy (system agency) through concurrent interactions that enable AI to function as a collaborator [15]. Concurrent interactions in group brainstorming support both higher human agency and AI autonomy during divergent and convergent thinking. In contrast to these creative product-focused systems, we seek understanding of user perceptions of human agency during the idea selection of the ideas themselves.

3 Method

3.1 Design Considerations

In group brainstorming, divergent thinking and convergent thinking are both sequentially and cyclically related. Thus, to explore different levels of human agency and AI autonomy in convergent thinking, we include both a chatroom and ideation map visualization as interaction modalities for idea generation and selection, respectively. To facilitate multiple iterations of convergent thinking, we opt to use an ideation map visualization to function as a shared knowledge workspace, inspired by post-it brainstorming with digital affordances. Ideas can be initially selected from text messages by clicking the respective "Post" button next to each message. The edit and delete modes enable users to modify ideas and filter them, and thus more idea selection iterations beyond the original "post" selections. To help users visualize similarities between ideas, we automatically cluster ideas based on their semantic similarities, connecting similar ideas together with edge lengths proportional to their semantic distances. In addition, to help users track who selected ideas, each idea's color in the ideation map reflects the respective poster's color displayed in the username sidebar (Figure 1).

We opt to have AI interact more like a human teammate to better match the human users' actions, and elicit feedback on levels of AI autonomy. To do so, we use a randomized sequential turn-taking order for the AI for two reasons: 1) the AI will perform actions more like a user, pausing in between actions similarly to a human teammate, and 2) enforcing that the AI performs the same number of actions as the average human user. The AI can either suggest a new idea (divergent thinking) or contribute to the initial idea

selection via posting a new idea (convergent thinking). Restricting the AI from editing and deleting ideas enables the human to retain full control over the final output, consistent with prior work on selection [2].

3.2 IdeaMap Implementation

IdeaMap was implemented as a Flask-based web app to provide cross-browser and device compatibility. Two instances of gpt-3.5-turbo chat completions were used: one for the AI collaborator, and the other for topic extraction used for posting an idea from a message onto the map. The prompt for the AI collaborator included the set of brainstorming guidelines given to the human participants with a few extra rules: to list exactly one idea, provide a 1-sentence explanation, use a maximum of 30 words, and provide either a related or different idea to the last inputted idea. The prompt for the topic extraction used few-shot prompting and asked to return the idea of an input sentence as a noun. The ideation map was implemented on the client side with d3-force. Server-side parameters were used to change the brainstorming task topic and control the frequency of GPT posts according to the different number of users in the single and multi-party probes.

3.3 Participants

We recruited 12 participants aged 23 to 36 years ($M = 26.25$, $SD = 3.93$), including 7 females and 5 males, from local university networks. Since participants needed to be able to communicate with both GPT and another participant during the group brainstorming task, we filtered for participants that are native or fluent in English. We also filtered for participants that have both previous experience with AI and an interest in using AI for knowledge work to collect user-as-designer informant perspectives. All participants gave informed consent before starting the workshop.

3.4 Participatory Design Workshop Procedure

Our remote participatory design workshop took approximately 90 minutes, in 4 phases: (1) System tutorial, (2) Group brainstorming task (technology probes), (3) Participatory design workshop, and (4) Demographic questionnaire. Each workshop session had 2 participants.

3.4.1 System Tutorial. Upon completing the consent form, participants were given 5 minutes to read through a system tutorial that demonstrates IdeaMap's functionality and ask questions about how the system works.

3.4.2 Technology Probes. Participants completed two design thinking sessions with an AI collaborator using IdeaMap as a technology probe [12]. The first 10-minute session was in a single-party human-AI team (1 human, 1 AI) and was followed by a 15-minute session in a multi-party human-AI team (2 humans, 1 AI). Each session used a different topic (either a desert or moon survival topic) assigned in a counterbalanced order; the tasks were inspired by the NASA moon survival team building exercise [21]. Groups were tasked with trying to brainstorm new ideas with IdeaMap, using the chatroom for idea generation (divergent thinking) and the ideation map visualization for idea selection (convergent thinking) as the final

brainstorming output. Participants were given a set of brainstorming rules for the chatroom based on prior work [29]: 1. the more ideas the better, 2. the wilder the ideas the better, 3. a combination and improvement of ideas are better, 4. avoid evaluating others' ideas, 5. only list one idea per text entry, 6. do not repeat ideas. Participants were also instructed to both post "good" ideas to the ideation map and filter ideas down to the best ones using the map edit and delete modes throughout the task sessions.

3.4.3 Participatory Design Workshop. The participatory design workshop took approximately 1 hour. Participants were given a shared Miro board template with questions for digital post-it note brainstorming (Figure 2). Topics focused on what features an AI collaborator should have and the level of AI involvement in idea selection within a group brainstorming context. Our goal was to learn more about user perceptions of human agency and AI autonomy, and user needs regarding how to best balance human agency and AI autonomy. Before the group brainstorming session, participants were given a few minutes to explore the Miro board and think about the questions. The group brainstorming session had a semi-structured format, in which we reviewed each question together.

3.4.4 Demographic Questionnaire. Participants completed a 1 minute demographic questionnaire about their age, gender, highest education level, and previous experience working with AI.

4 Results and Interpretations

We compiled the Miro boards from all of the participatory design workshop sessions, and used affinity diagrams to derive themes about participants' perceptions of human agency and AI autonomy [16].

4.1 Perceived Human Agency and AI Autonomy in Convergent Thinking (RQ 1)

Humans have indicated a preference for user control over the final output of human-AI teaming tasks [1, 26]. Consistent with this prior work, we found that participants not only sought control over the final output during human-AI idea selection, but also expressed several reasons for this desire to override AI autonomy. Doubts about AI's capabilities to make good idea selections led to conclusions that AI "doesn't seem smart enough for complex tasks" (P2, P7). This perceived incompetence, combined with participants claiming that they "don't know how much to trust AI" (P8, P11), suggests a preference for human control during co-selection tasks. However, for an AI to be considered an equal teammate in a human-AI team, the AI also needs its own autonomy during idea selection. One way that humans could have greater human control while retaining AI autonomy is via multi-party groups. Participants seem to prefer working in multi-party groups, citing the ability to have "human verification of idea selection" (P5, P6). Another way that humans could have more control over the human-AI team is by having a more passively involved AI collaborator. We found that a majority of participants prefer a more passive AI collaborator for idea selection in both single-party teams (P3-P9, P11-P12) and multi-party teams (P1, P3-P5, P7-P8, P11-P12). Cited reasons for this preference include AI's lack of transparency and inability to

explain why it's making a selection (P7, P8). Ultimately, participants seemed to judge AI's capabilities based on its demonstrated lack of reliability and trustworthiness, as suggested by Shneiderman's RST framework [25] and prior work in the healthcare domain [2].

Despite participants' preference for a more passive AI collaborator for idea co-selection, participants acknowledge the potential benefits of having AI help them in earlier iteration stages of idea selection. AI's computational strengths could help with organizational efficiency, from "[helping] the user find key terms" for faster iteration (P2) to providing automated filtering and consolidation of similar ideas (P1-P4). This could be particularly helpful when dealing with large amounts of data: P12 notes "when there are many, many people (e.g., 1000), AI can help filter/combine ideas to a human-searchable number." In addition, AI also has the potential to play several types of supporting roles with greater autonomy. For example, when AI selected a human-generated idea and when participants selected an AI-generated idea in the first iteration of idea selection, P10 felt more confident in their ideas, as if they "were agreeing with an AI idea or vice versa." This validation could help human teammates feel more satisfied with the co-selection results. Furthermore, several participants noted situations throughout the idea selection process in which AI could serve as a mediator in multi-party teams (P7-P9, P12). Having a built-in mediator could help resolve disagreements about the best idea faster (P7-P8), particularly with an "unbiased judge of ideas" (P9). Suitable applications of these AI supporting roles, and thus AI autonomy, seems to be situationally dependent: thus, people could benefit from different levels of AI autonomy corresponding to different idea selection iteration stages.

While participants discussed their perceptions of AI broadly, and not specifically the gpt-3.5-turbo AI used in our IdeaMap technology probe, their responses were still influenced by our technology probe. In future work, a combination of other more advanced LLMs like gpt-4-turbo and additional prompt engineering could potentially create an AI that can provide better idea selections and explanations. This improved task performance could affect users' willingness to rely and trust AI, resulting in different user perceptions of desired human agency and AI autonomy.

4.2 Situationally Aware AI (RQ 2)

Variance in participant performance throughout an idea selection task suggests a need for a flexible, situationally aware AI that can seamlessly change its behavior (and thus, levels of AI autonomy) depending on current human behaviors. This situational awareness—AI's ability to collect input data on user teammate behavior, analyze it, and alter its own behavior (levels of AI autonomy) accordingly [6]—can impact team performance. There are two ways that AI can gather user behavior data: explicitly via human input, and implicitly through AI's own observations of current user-specific behaviors. In parallel, humans need to exhibit a similar awareness of both other humans' and AI's behavior by monitoring and responding to explicit and implicit human and AI social cues. While this awareness in human-only groups has been established in prior work on social translucence [7] and grounding in communication [3], future work could explore how grounding can be created between human and AI teammates.

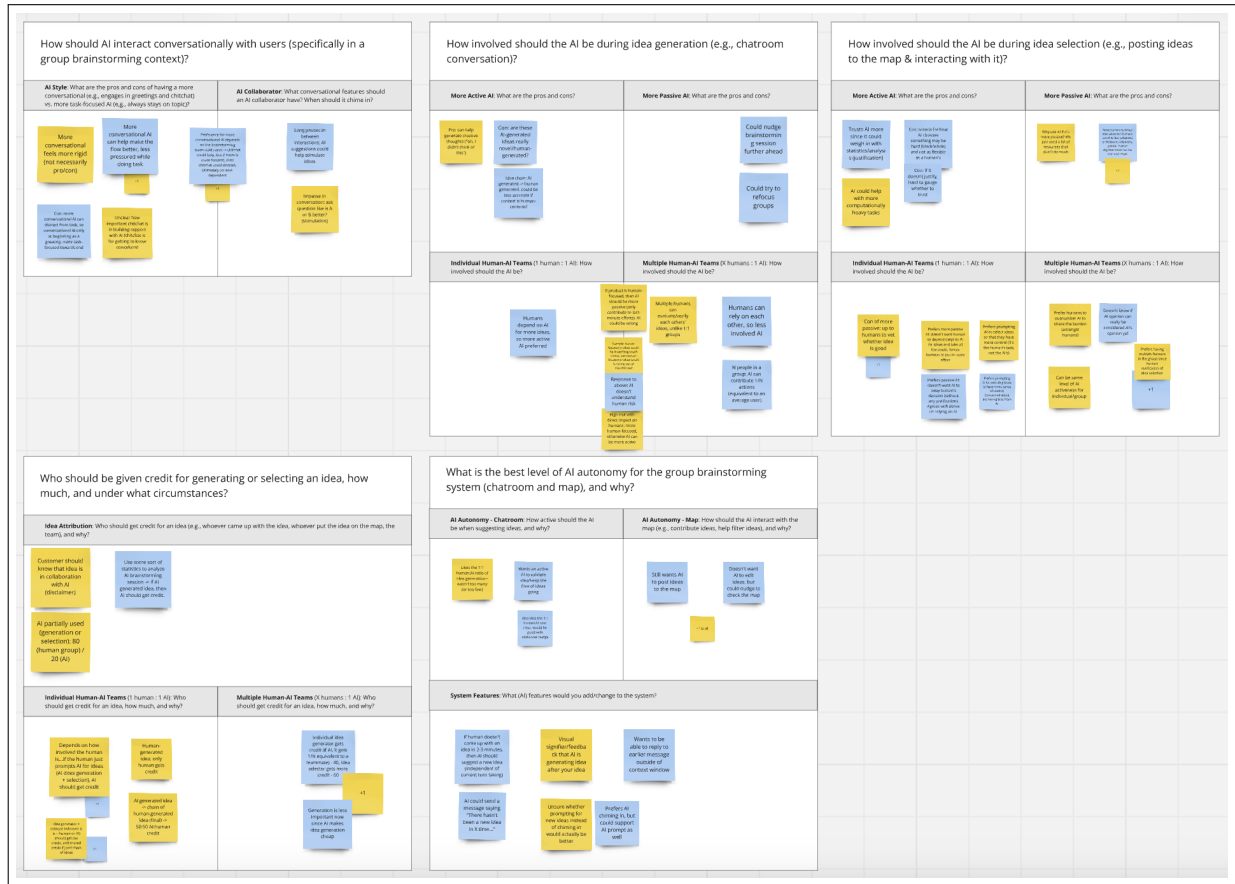


Figure 2: Miro board from a participatory design workshop session.

A situationally aware AI that adjusts its behavior according to explicit human feedback enables it to retain autonomy as a teammate while aligning with users' expectations of AI behavior. This is similar to how humans often adjust their performance based on constructive feedback from others. Participants express their desire for such explicit feedback ranging from "[choosing] to let AI generate more ideas or select/critique previous ideas" (P3-P4) to "[giving] AI feedback (e.g., more realistic or more creative ideas) via thumbs up/down" (P8). Ultimately, this would enable humans to retain control over when and how the balance of human agency and AI autonomy shifts.

There are also situations in which participants deem it appropriate for AI to exert greater autonomy and automatically shift the balance of human agency and AI autonomy according to observed human behavior. Participants suggest that a flexible AI that adapts its behavior in these scenarios could potentially boost team performance. For example, P10 suggests that the AI could "[become] more active if the humans are inactive," and P2 notes that the AI could "gauge how many ideas the users are posting and try to be of equal contribution." Human-AI teams with a situationally aware AI that responds to both explicit and these implicitly observed human social cues have the potential to generate better human-AI synergy support through better balanced team actions and optimize for overall task performance.

5 Conclusion

We explored the usage of IdeaMap, a human-AI brainstorming system with a chatroom and ideation map visualization as a technology probe for human agency and AI autonomy through a participatory design workshop. We identified user perceptions of human agency and AI autonomy in idea selection: a desire for high levels of human control and lower levels of AI autonomy, attributed to doubts about AI's capabilities, trust, and transparency (RQ1). Furthermore, we examined the balance of human agency and AI autonomy over idea selection in a shared knowledge workspace, resulting in an AI designed with situational awareness (RQ2). This flexible AI seamlessly adjusts its own actions based on individual user behaviors, through both direct user feedback and its own observations of user behavior. Future work includes exploring different components of situationally aware AI and its effects on multi-party group dynamics, such as human-AI team synergy.

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References

- [1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N Bennett, Kori Inkpen, et al. 2019. Guidelines for human-AI interaction. In *Proceedings of the 2019 chi conference on human factors in computing systems*. 1–13.
- [2] Onur Asan, Alparslan Emrah Bayrak, Avishkek Choudhury, et al. 2020. Artificial intelligence and human trust in healthcare: focus on clinicians. *Journal of medical Internet research* 22, 6 (2020), e15154.
- [3] Herbert Clark and Susan Brennan. 1991. Grounding in communication. *Perspectives on socially shared cognition* (1991).
- [4] Design Council. 2025. The Double Diamond. <https://www.designcouncil.org.uk/our-resources/the-double-diamond/>
- [5] Rikke Friis Dam. 2025. 5 Stages in the Design Thinking Process. <https://www.designcouncil.org.uk/our-resources/the-double-diamond/>
- [6] Mica R Endsley. 2023. Supporting Human-AI Teams: Transparency, explainability, and situation awareness. *Computers in Human Behavior* 140 (2023), 107574.
- [7] Thomas Erickson and Wendy A Kellogg. 2000. Social translucence: an approach to designing systems that support social processes. *ACM transactions on computer-human interaction (TOCHI)* 7, 1 (2000), 59–83.
- [8] Xianzhe Fan, Zihan Wu, Chun Yu, Fenggui Rao, Weinan Shi, and Teng Tu. 2024. ContextCam: Bridging Context Awareness with Creative Human-AI Image Co-Creation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–17.
- [9] Jonathan Grudin. 2017. *From Tool to Partner: The Evolution of Human-Computer Interaction* (1 ed.). Springer Cham.
- [10] Jiajing Guo, Vikram Mohanty, Jorge H Piazzentin Ono, Hongtao Hao, Liang Gou, and Liu Ren. 2024. Investigating Interaction Modes and User Agency in Human-LLM Collaboration for Domain-Specific Data Analysis. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–9.
- [11] Yuanning Han, Ziyi Qiu, Jiale Cheng, and RAY LC. 2024. When Teams Embrace AI: Human Collaboration Strategies in Generative Prompting in a Creative Design Task. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–14.
- [12] Hilary Hutchinson, Wendy Mackay, Bo Westerlund, Benjamin B Bederson, Alison Druin, Catherine Plaisant, Michel Beaudouin-Lafon, Stéphane Conversy, Helen Evans, Heiko Hansen, et al. 2003. Technology probes: inspiring design for and with families. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 17–24.
- [13] Jeongyeon Kim, Sangho Suh, Lydia B Chilton, and Haijun Xia. 2023. Metaphorian: Leveraging Large Language Models to Support Extended Metaphor Creation for Science Writing. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 115–135.
- [14] Vivian Lai, Chacha Chen, Alison Smith-Renner, Q Vera Liao, and Chenhao Tan. 2023. Towards a science of human-ai decision making: An overview of design space in empirical human-subject studies. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. 1369–1385.
- [15] Tomas Lawton, Kazjon Grace, and Francisco J Ibarrola. 2023. When is a Tool a Tool? User Perceptions of System Agency in Human-AI Co-Creative Drawing. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 1978–1996.
- [16] Andrés Lucero. 2015. Using affinity diagrams to evaluate interactive prototypes. In *Human-Computer Interaction—INTERACT 2015: 15th IFIP TC 13 International Conference, Bamberg, Germany, September 14–18, 2015, Proceedings, Part II* 15. Springer, 231–248.
- [17] Microsoft. 2025. Microsoft Copilot. <https://copilot.microsoft.com/>
- [18] Caterina Moruzzi and Solange Margarido. 2024. A user-centered framework for human-ai co-creativity. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–9.
- [19] Michael Muller and Justin Weisz. 2022. Extending a human-ai collaboration framework with dynamism and sociality. In *Proceedings of the 1st Annual Meeting of the Symposium on Human-Computer Interaction for Work*. 1–12.
- [20] Imani Munyaka, Zahra Ashktorab, Casey Dugan, James Johnson, and Qian Pan. 2023. Decision making strategies and team efficacy in human-AI teams. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW1 (2023), 1–24.
- [21] NASA. 2016. NASA Exercise: Ranking Survival Objects for the Moon. <https://www.psychologicalscience.org/observer/nasa-exercise>
- [22] Vildan Salikutluk, Janik Schöpper, Franziska Herbert, Katrin Scheuermann, Eric Frodl, Dirk Balfanz, Frank Jäkel, and Dorothea Koert. 2024. An Evaluation of Situational Autonomy for Human-AI Collaboration in a Shared Workspace Setting. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–17.
- [23] Orit Shaer, Angelora Cooper, Osnat Mokryn, Andrew L Kun, and Hagit Ben Shoshan. 2024. AI-Augmented Brainwriting: Investigating the use of LLMs in group ideation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–17.
- [24] Donghoon Shin, Soomin Kim, Ruoxi Shang, Joonhwan Lee, and Gary Hsieh. 2023. IntroBot: Exploring the use of chatbot-assisted familiarization in online collaborative groups. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [25] Ben Shneiderman. 2020. Human-centered artificial intelligence: Reliable, safe & trustworthy. *International Journal of Human-Computer Interaction* 36, 6 (2020), 495–504.
- [26] Ben Shneiderman and Catherine Plaisant. 2010. *Designing the user interface: strategies for effective human-computer interaction*. Pearson Education India.
- [27] Sangho Suh, Meng Chen, Bryan Min, Toby Jia-Jun Li, and Haijun Xia. 2024. Luminate: Structured Generation and Exploration of Design Space with Large Language Models for Human-AI Co-Creation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–26.
- [28] Qian Wan and Zhicong Lu. 2023. GANCollage: A GAN-Driven Digital Mood Board to Facilitate Ideation in Creativity Support. In *Proceedings of the 2023 ACM Designing Interactive Systems Conference*. 136–146.
- [29] Hao-Chuan Wang, Susan R Fussell, and Dan Cosley. 2011. From diversity to creativity: Stimulating group brainstorming with cultural differences and conversationally-retrieved pictures. In *Proceedings of the ACM 2011 conference on Computer supported cooperative work*. 265–274.
- [30] Sitong Wang, Samia Menon, Tao Long, Keren Henderson, Dingzeyu Li, Kevin Crowston, Mark Hansen, Jeffrey V Nickerson, and Lydia B Chilton. 2024. Reel-Framer: Human-AI co-creation for news-to-video translation. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–20.
- [31] Chengbo Zheng, Yuheng Wu, Chuhan Shi, Shuai Ma, Jiehui Luo, and Xiaojuan Ma. 2023. Competent but Rigid: Identifying the Gap in Empowering AI to Participate Equally in Group Decision-Making. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–19.